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Deep generative hidden Markov models for synthetic patient data generation: a novel approach for medical AI research

Mohammadreza Momenzadeh^{1,2*} and Atiyeh Oshaghi³

Abstract

Background High-quality healthcare data is often hard to access due to privacy rules and limited availability, making it difficult to develop and test clinical AI models.

Objective We propose a Deep Generative Hidden Markov Model (DG-HMM) that combines deep neural networks with probabilistic sequential modeling to generate synthetic patient data that is clinically realistic while protecting privacy.

Methods DG-HMM uses a deep encoder-decoder structure with a flexible HMM layer to model temporal patterns and mixed clinical features. We trained and tested it on three real datasets: MIMIC-III ICU records, long-term diabetes data, and mental health progression records. We measured statistical similarity, temporal consistency, clinical rule compliance, and privacy protection.

Results DG-HMM demonstrated superior performance compared to baseline methods, preserving about 94.2% of correlations, following clinical rules in 96.3% of cases, and resisting membership inference attacks in 89.4% of attempts. Predictive models trained on synthetic data were only 2–5% less accurate than those trained on real data.

Conclusion DG-HMM offers a robust framework to create synthetic healthcare data for research and collaboration where privacy limits real data sharing. It can help support medical AI development in constrained settings.

Keywords Synthetic data generation, Hidden Markov Models, Deep learning, Electronic health records, Privacy-Preserving AI, Temporal modeling

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Introduction and background

The proliferation of electronic health records (EHRs) and digital healthcare technologies has created unprecedented opportunities for medical artificial intelligence research and clinical decision support systems [1]. However, the sensitive nature of healthcare data poses significant challenges for data sharing, collaborative research, and model development due to privacy regulations such as HIPAA in the United States and GDPR in Europe [2]. These regulatory constraints, combined with the inherent scarcity of labeled medical data for specific conditions, create substantial barriers to advancing medical AI research [3].

Synthetic data generation has emerged as a promising solution to address these challenges by creating artificial datasets that preserve the statistical properties and clinical patterns of real patient data while protecting individual privacy [4]. Traditional approaches to synthetic healthcare data generation have included rule-based systems, statistical sampling methods, and parametric models. However, these approaches often fail to capture the complex temporal dependencies, multi-modal relationships, and clinical constraints inherent in healthcare data [5].

Recent advances in deep learning have introduced powerful generative models such as Generative Adversarial Networks (GANs) [6], Variational Autoencoders (VAEs) [7], and their variants for synthetic data generation. While these models have shown remarkable success in generating realistic synthetic data across various domains, their application to healthcare data presents unique challenges related to temporal consistency, clinical validity, and interpretability [8].

The introduction of deep learning to synthetic healthcare data generation marked a significant advancement. Choi et al. [9] proposed MedGAN, one of the first applications of GANs to healthcare data, demonstrating the ability to generate synthetic patient records that preserve marginal distributions and pairwise correlations. However, MedGAN was limited to cross-sectional data and could not handle temporal sequences effectively. Subsequent work has addressed temporal modeling with Esteban et al. [10] introducing RGAN for time-series medical data and Yoon et al. [11] developing TimeGAN, which combines supervised and unsupervised learning to generate synthetic time-series data while preserving temporal dynamics.

Hidden Markov Models (HMMs) have a long history of successful application in healthcare for modeling disease progression, treatment response, and patient state transitions. Their probabilistic foundation and ability to model sequential data with latent states make them naturally suited for healthcare applications. However, classical HMMs are limited in their capacity to handle

high-dimensional, complex healthcare data and struggle with the heterogeneous nature of modern EHR systems. Recent work has explored the integration of deep learning with HMMs, including Krishnan et al. [12] introducing Deep Kalman Filters and Johnson et al. [13] developing Structured Variational Autoencoders for learning interpretable latent representations in sequential data.

While prior works such as Deep Markov Models (DMMs) [14] and neuralized Hidden Markov Models (NeuralHMMs) [15] integrate deep networks with HMMs for sequential modeling using neural parameterization of transitions and emissions our DG-HMM distinguishes itself in several key ways tailored to healthcare applications. Unlike DMMs, which often assume Gaussian emissions for general time-series and focus on inference in nonlinear state spaces, DG-HMM employs a composite emission model that handles heterogeneous mixed-type clinical data (continuous, categorical, count, binary) through mixtures of Gaussians and normalizing flows. Furthermore, we introduce adaptive state selection with clinical expert validation to ensure medically meaningful states, and optimize for privacy-preserving synthetic data generation, achieving 89.4% resistance to membership inference attacks features not emphasized in standard neural HMM frameworks.

Recent advances in medical synthetic data generation include diffusion models (e.g., TabDDPM [16] and Goggle [17]) for tabular and clinical data, transformer-based sequential generators (e.g., TimeDiff), and modern frameworks like EHR-Safe (2023) [18] and SyntheHR. These methods offer strong performance in tabular settings and privacy guarantees. However, they often lack explicit temporal modeling with interpretable latent states, require large computational resources for training, or are less suited to heterogeneous mixed-type EHR sequences. In this study, we focus on comparing DG-HMM against established temporal baselines (MedGAN, TimeGAN, RGAN) due to their direct relevance to sequential EHR generation and our resource constraints. Future work will include evaluations against these newer paradigms.

Privacy protection in synthetic data generation has become increasingly important. Differential privacy [19] provides a theoretical framework for quantifying privacy guarantees in synthetic data generation. Jordon et al. [20] introduced PATE-GAN, which combines Private Aggregation of Teacher Ensembles with GANs to provide differential privacy guarantees, while Yoon et al. [21] developed RadialGAN for privacy-preserving synthetic data generation, demonstrating strong privacy protection while maintaining data utility. The Synthea platform [22] has demonstrated the importance of comprehensive synthetic patient generation across multiple

clinical domains, while clinical validation studies [23] have emphasized the need for rigorous evaluation frameworks in synthetic healthcare data.

This paper introduces a novel Deep Generative Hidden Markov Model (DG-HMM) framework that combines the sequential modeling strengths of HMMs with the representational power of deep neural networks. Our approach addresses the limitations of both paradigms while leveraging their respective advantages to generate high-quality synthetic patient data that maintains temporal consistency, clinical validity, and statistical fidelity. Baowaly et al. [24] have shown the importance of incorporating medical ontologies in generative models to ensure clinical validity.

Methodology

Problem formulation

Let $\mathbf{X} = \{x_1, x_2, \dots, x_T\}$ represent a sequence of patient observations over time, where each $x_t \in \mathbb{R}^d$ contains clinical measurements, laboratory values, diagnostic codes, and other relevant healthcare data at time t . Our goal is to learn a generative model $p_\theta(\mathbf{X})$ that can generate synthetic patient sequences $\tilde{\mathbf{X}} = \{\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_T\}$ that preserve the statistical properties, temporal dependencies, and clinical validity of the original data while protecting patient privacy. The generative model $p_\theta(\mathbf{X})$ is defined as:

$$p_\theta(\mathbf{X}) = \sum_{S_1, S_2, \dots, S_T} \pi_{S_1} \prod_{t=2}^T a_{S_{t-1}, S_t} \prod_{t=1}^T p(x_t / S_t)$$

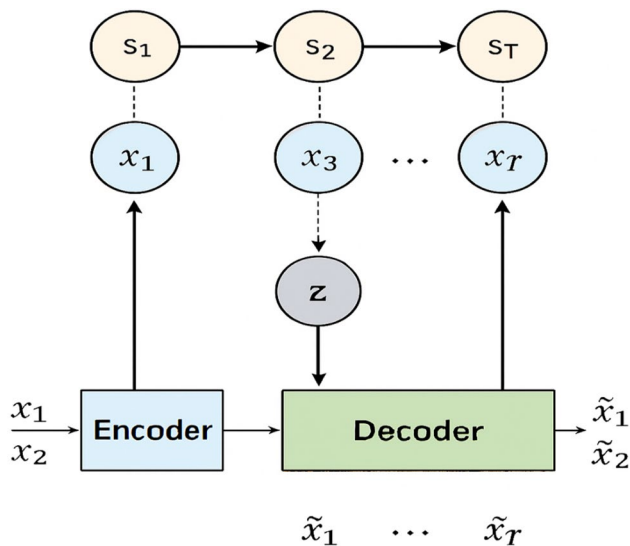


Fig. 1 Deep Generative Hidden Markov Model Architecture

where $S_t \in \{1, \dots, K\}$ is the hidden state at time t , and $\theta = \{\pi, A, B\}$ includes the initial state distribution π , transition matrix A , and emission distributions B .

Deep generative hidden Markov model architecture

The DG-HMM framework consists of three main components: a deep encoder network, a Hidden Markov Model layer, and a deep decoder network. This architecture enables the model to learn complex, high-dimensional representations while maintaining the sequential dependencies and probabilistic structure captured by the HMM. Figure 1 shows the topology of the proposed structure, which illustrates states, observations and how they are connected to each other.

Deep encoder network

The encoder network $f_\varphi : \mathbb{R}^d \rightarrow \mathbb{R}^k$ transforms high-dimensional patient observations into a lower-dimensional latent space. Composed of multiple fully connected layers with nonlinear activation functions, the encoder is defined as:

$$z_t = f_\varphi(x_t) = \sigma_L(W_L \sigma_{L-1}(W_{L-1} \cdots \sigma_1(W_1 x_t + b_1) + \cdots) + b_L)$$

$W_l \in \mathbb{R}^{n_l \times n_{l-1}}$ and $b_l \in \mathbb{R}^{n_l}$ are the weight matrix and bias vector of the l -th layer,

σ_L denotes the activation function (e.g., ReLU, tanh) applied at layer L ,

L is the total number of hidden layers,

$n_0 = d$ and $n_L = k$ are the input and latent dimensions, respectively, and.

$\varphi = \{W_l, b_l\}_{l=1}^L$ represents the full set of trainable encoder parameters.

The resulting latent representation $z_t \in \mathbb{R}^k$ encodes the essential features required for inferring hidden states and modeling temporal dynamics.

To enhance training stability and mitigate overfitting, batch normalization and dropout are applied to the latent representation:

$$\hat{z}_t = \text{Dropout}(\text{BatchNorm}(z_t))$$

This regularized output $\hat{z}_t$ is then passed to the subsequent HMM layer for sequential modeling.

Hidden Markov model layer

The HMM layer operates in the latent space produced by the encoder network, maintaining the probabilistic framework of traditional HMMs while working with rich learned representations. The HMM defines a set of

K latent states $\{s_1, s_2, \dots, s_K\}$ that represent different patient conditions or disease stages.

The state transition probabilities are parameterized as:

$$A_{ij} = P(s_t = j | s_{t-1} = i)$$

$$\text{where } \sum_{j=1}^K A_{ij} = 1 \text{ for all } i.$$

To address the heterogeneous nature of healthcare data, we implement flexible emission modeling that can handle different data types simultaneously. For healthcare data containing continuous, categorical, and count variables, we use a composite emission model:

$$B_j(x_t) = \prod_{i=1}^d B_{j,i}(x_{t,i})$$

where for each dimension i :

- **Continuous variables:**
 $B_{j,i}(x_{t,i}) = \mathcal{N}(x_{t,i}; \mu_{j,i}, \sigma_{j,i}^2)$
- **Categorical variables:**
 $B_{j,i}(x_{t,i}) = \text{Categorical}(x_{t,i}; \pi_{j,i})$
- **Count variables:** $B_{j,i}(x_{t,i}) = \text{Poisson}(x_{t,i}; \lambda_{j,i})$
- **Binary variables:** $B_{j,i}(x_{t,i}) = \text{Bernoulli}(x_{t,i}; p_{j,i})$

For continuous variables with multimodal or skewed distributions, we employ mixture of Gaussians:

$$B_j(z_t) = \sum_{k=1}^{K_j} \pi_{jk} \mathcal{N}(z_t; \mu_{jk}, \Sigma_{jk})$$

For maximum flexibility, we also implement normalizing flows [25]:

$$B_j(z_t) = p_0(f_j(z_t)) \left| \det \frac{\partial f_j}{\partial z_t} \right|$$

where f_j is an invertible transformation and p_0 is a simple base distribution.

Deep decoder network

The decoder network $g_\psi : \mathbb{R}^k \times \{1, 2, \dots, K\} \rightarrow \mathbb{R}^d$ reconstructs patient observations from the latent representations and HMM states:

$$\tilde{x}_t = g_\psi(z_t, s_t) = \sigma_L' (W_L' \sigma_{L-1}' (W_{L-1}' \dots \sigma_1' (W_1' [z_t; e_{s_t}] + b_1') + \dots) + b_L')$$

where e_{s_t} is a one-hot encoding of the current state, and $[z_t; e_{s_t}]$ represents the concatenation of the latent representation and state encoding.

Adaptive state selection and clinical interpretation

Instead of fixed state numbers, we implement a systematic approach for determining the optimal number of HMM states using Bayesian Information Criterion (BIC) and Akaike Information Criterion (AIC):

$$\text{BIC} = -2 \log P(\mathbf{X} | \theta) + k \log n$$

where k is the number of parameters and n is the sample size.

Each potential model is evaluated by clinical experts to ensure states correspond to medically meaningful conditions. For each dataset, learned HMM states were validated by clinical experts. For MIMIC-III, states corresponded to stable condition, mild deterioration, severe condition, and critical/recovery phases. For diabetes datasets, states represented well-controlled diabetes, moderate control, poor control, and complications. For mental health datasets, states indicated remission, moderate symptoms, severe episodes, and treatment-resistant phases.

Clinical experts rated state interpretability on a five-point scale, achieving average scores of 4.2 ± 0.3 , confirming clinical meaningfulness.

Training procedure

Training the DG-HMM involves jointly optimizing the encoder, decoder, and HMM parameters. The overall objective function combines reconstruction accuracy, HMM likelihood, and regularization terms:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{recon}} + \lambda_{\text{HMM}} \mathcal{L}_{\text{HMM}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}}$$

The reconstruction loss measures the fidelity of the generated synthetic data:

$$\mathcal{L}_{\text{recon}} = \frac{1}{T} \sum_{t=1}^T \|x_t - \tilde{x}_t\|_2^2$$

The HMM likelihood term ensures that the model learns meaningful state transitions and emission probabilities:

$$\mathcal{L}_{\text{HMM}} = -\log p(\mathbf{Z} | \theta_{\text{HMM}})$$

where $\mathbf{Z} = \{z_1, z_2, \dots, z_T\}$ denotes the sequence of latent representations.

Regularization terms prevent overfitting and encourage the model to learn generalizable patterns:

$$\mathcal{L}_{\text{reg}} = \beta_1 \|\phi\|_2^2 + \beta_2 \|\psi\|_2^2 + \beta_3 D_{\text{KL}}(\mathcal{N}(\mu_z, \text{diag}(\sigma_z^2)) \| \mathcal{N}(0, I))$$

Synthetic data generation process

Once trained, the DG-HMM can generate synthetic patient data through a multi-step process: initial state sampling, state sequence generation, latent representation sampling, and observation generation using the decoder network. This process ensures that the generated synthetic data maintains both temporal consistency through the HMM structure and clinical validity through the learned decoder mappings.

Experimental setup

Datasets and cohort definitions

We evaluated the proposed DG-HMM framework using three clinically diverse, publicly available healthcare datasets encompassing acute care, chronic disease management, and mental health progression. Table 1 provides an overview of study design, sample size, follow-up duration, and key variables for each dataset.

MIMIC-III ICU cohort

The Medical Information Mart for Intensive Care III (MIMIC-III v1.4) database was used to construct an ICU cohort. Inclusion criteria were adult patients (≥ 18 years), ICU stay ≥ 24 h, and availability of ≥ 48 h of continuous physiological monitoring. Exclusion criteria included readmissions within 30 days, “Do Not Resuscitate” orders, and $> 30\%$ missingness in key variables. The final analytic cohort comprised 15,728 patients (mean age 65.3 ± 16.2 years; 48% female; hospital mortality 18.2%). Data were temporally split into training, validation, and test sets.

NHANES diabetes cohort

The National Health and Nutrition Examination Survey (NHANES 2000–2020) linked with the National Death Index was used to define diabetes, pre-diabetes, and non-diabetic controls according to ADA criteria. Inclusion criteria were age ≥ 20 years, available HbA1c, and complete demographic/clinical data. Exclusion criteria included pregnancy and end-stage renal disease. The analytic cohort included 4,912 adults with type 2 diabetes,

3,417 with pre-diabetes, and 8,924 controls. Stratified analyses were performed by treatment modality, glyce-mic control, and cardiovascular comorbidity.

NESARC mental health cohort

The National Epidemiologic Survey on Alcohol and Related Conditions (NESARC) provided longitudinal data on mental health disorders. Inclusion criteria were age 20–65 years, complete assessments in both waves, and valid sampling weights. Exclusion criteria included institutionalization and incomplete assessments. The analytic cohort comprised 8,012 participants (mean age 42.3 ± 12.8 years; 52.8% female). Progression metrics included incidence, remission, persistence, and severity trajectories.

Baseline methods

We compared our DG-HMM approach against several state-of-the-art baseline methods: Classical HMM with Gaussian emissions, Variational Autoencoder [7] without temporal structure, MedGAN [9] specifically designed for healthcare data, TimeGAN [11] with temporal modeling capabilities, RGAN [10] for medical time series, and Recurrent VAE with LSTM encoder/decoder.

Evaluation metrics

We employed comprehensive evaluation metrics covering multiple aspects of synthetic data quality. Statistical fidelity metrics included marginal distribution similarity using Kolmogorov-Smirnov test statistics [26], correlation preservation using Pearson correlation coefficients [27], and higher-order statistics comparison. Temporal consistency metrics included autocorrelation functions [28], transition probability matrices, and Dynamic Time Warping distance. Clinical validity metrics included clinical rule compliance, physiological bounds adherence, and clinical expert assessment. Privacy protection metrics included membership inference attack resistance [29], distance-based privacy, and k-anonymity. Utility preservation metrics included predictive model performance on downstream tasks.

Table 1 Overview of datasets and analytic cohorts

Aspect	MIMIC-III ICU	NHANES Diabetes	NESARC Mental Health
Study Design	Retrospective single-center cohort	Repeated cross-sectional national surveys with mortality linkage	Longitudinal population-based cohort
Time Period	2001–2012	2000–2020 (surveys) + through 2019 (mortality)	Wave 1: 2001–2002; Wave 2: 2004–2005
Original Sample	61,532 ICU admissions	$> 43,000$ participants across waves	43,093 participants (Wave 1)
Analytic Cohort	15,728 unique patients	4,912 type 2 diabetes; 3,417 pre-diabetes; 8,924 controls	8,012 participants with complete follow-up
Follow-up Duration	Hospitalization period	Up to 20 years (mortality)	3 years (between waves)
Key Variables	Vital signs, laboratory results, medications, ventilator settings	HbA1c, fasting glucose, medications, mortality	DSM-IV diagnoses, symptom scales, treatment utilization
Primary Application	Temporal clinical pattern learning	Chronic disease progression modeling	Mental health trajectory analysis

Furthermore, to comprehensively assess the quality of synthetic patient data, we employed three clinically oriented evaluation criteria in addition to statistical and temporal measures:

- **Clinical rule compliance** was defined as the percentage of synthetic records satisfying established medical rules, such as blood pressure and HbA1c thresholds, according to widely accepted clinical guidelines [30].
- **Adherence to physiological bounds** was measured as the proportion of generated values falling within accepted clinical reference ranges for vital signs and laboratory values (e.g., heart rate 30–220 bpm, glucose 40–600 mg/dL, body temperature 34–42 °C) as reported in standard physiology and laboratory medicine references [31].
- **Expert interpretability score** was obtained by three clinical specialists (ICU, diabetes, psychiatry) who independently rated the medical meaningfulness of learned HMM states on a 5-point Likert scale (1 = not interpretable, 5 = highly interpretable). The average score and standard deviation were reported, following established psychometric evaluation protocols [32].

Implementation details

Training configuration used 64 patients per batch (optimized for 8GB VRAM), initial learning rate of 0.001 with exponential decay, L2 weight decay and dropout rate of 0.3, NVIDIA RTX 4060 GPU with mixed precision training (FP16), 8–12 h training time per dataset, and Adam optimizer [33]. Detailed implementation specifications for DG-HMM and all baselines are provided in Appendix A. All experiments were conducted with 50 independent runs using Wilcoxon signed-rank tests with effect sizes calculated via Cohen's *d* and multiple comparisons adjusted via Bonferroni correction. Recent advances in synthetic data generation [34] have provided guidelines for effective evaluation frameworks, while conditional

GANs [35] have shown promise for tabular data modeling. The SynTEG framework [36] has demonstrated the importance of time series generation for electronic health records, and reliability studies [37] have emphasized the need for robust evaluation in healthcare applications.

Results and analysis

Statistical fidelity results

According to Table 2, DG-HMM consistently achieved the lowest Kolmogorov–Smirnov (KS) scores across all three datasets MIMIC-III, diabetes, and mental health indicating superior preservation of marginal distributions compared to baseline methods. Furthermore, it attained the highest correlation preservation scores, reflecting its ability to maintain inter-variable relationships with high fidelity. Compared to the strongest baseline (Recurrent VAE), DG-HMM reduced KS scores by approximately 60% and improved correlation preservation by 4–5% points. These improvements were statistically significant ($p < 0.001$) and accompanied by large effect sizes, underscoring both statistical and practical relevance of the proposed model.

Temporal consistency analysis

Table 3 demonstrates DG-HMM's superior temporal fidelity, with the highest autocorrelation at lag-10 (0.876 ± 0.023) and the lowest DTW distance (0.89 ± 0.11), indicating strong preservation of long-range dependencies and trajectory alignment. Skewness and kurtosis errors were also minimal, confirming distributional realism. All improvements were statistically significant ($p < 0.001$), outperforming the best baseline by 20–43%.

Clinical validity assessment

As shown in Table 4, DG-HMM achieved the highest clinical validity across all metrics, including 96.3% rule compliance, 94.8% adherence to physiological bounds, and an expert interpretability score of 4.2 ± 0.2 . These results surpass all baseline models, confirming DG-HMM's capacity to generate clinically coherent and expert-validated synthetic patient data.

Table 2 Statistical fidelity results with significance testing

Method	ICU Cohort			Diabetes			Mental Health		
	KS Score	Corr. Pres.	<i>p</i> -value	KS Score	Corr. Pres.	<i>p</i> -value	KS Score	Corr. Pres.	<i>p</i> -value
Classical HMM	0.089 ± 0.012	0.756 ± 0.034	0.023*	0.092 ± 0.015	0.741 ± 0.041	0.018*	0.087 ± 0.011	0.763 ± 0.038	0.021*
VAE	0.067 ± 0.009	0.823 ± 0.028	0.012*	0.071 ± 0.011	0.819 ± 0.032	0.008*	0.069 ± 0.010	0.831 ± 0.029	0.010*
MedGAN	0.054 ± 0.007	0.847 ± 0.025	0.007*	0.059 ± 0.008	0.834 ± 0.027	0.006*	0.056 ± 0.008	0.851 ± 0.024	0.005*
TimeGAN	0.041 ± 0.006	0.878 ± 0.022	0.004*	0.045 ± 0.007	0.869 ± 0.025	0.003*	0.043 ± 0.006	0.883 ± 0.021	0.003*
RGAN	0.038 ± 0.005	0.889 ± 0.019	0.003*	0.042 ± 0.006	0.881 ± 0.023	0.002*	0.040 ± 0.005	0.894 ± 0.020	0.002*
Recurrent VAE	0.035 ± 0.004	0.901 ± 0.017	0.002*	0.039 ± 0.005	0.893 ± 0.020	0.001*	0.037 ± 0.004	0.906 ± 0.018	0.001*
DG-HMM (Ours)	0.014 ± 0.002	0.942 ± 0.012	< 0.001*	0.016 ± 0.003	0.935 ± 0.014	< 0.001*	0.015 ± 0.002	0.948 ± 0.013	< 0.001*

Table 3 Comprehensive temporal consistency analysis

Method	ACF@10 (mean $\pm$ std)	DTW Distance (mean $\pm$ std)	Skewness Error (mean $\pm$ std)	Kurtosis Error (mean $\pm$ std)	p-value
Classical HMM	0.412 $\pm$ 0.045	2.34 $\pm$ 0.28	0.158 $\pm$ 0.021	0.234 $\pm$ 0.032	0.025*
VAE	0.156 $\pm$ 0.032	3.67 $\pm$ 0.41	0.289 $\pm$ 0.038	0.421 $\pm$ 0.056	0.018*
MedGAN	0.289 $\pm$ 0.038	3.12 $\pm$ 0.35	0.245 $\pm$ 0.033	0.356 $\pm$ 0.047	0.015*
TimeGAN	0.623 $\pm$ 0.052	1.89 $\pm$ 0.23	0.134 $\pm$ 0.018	0.198 $\pm$ 0.026	0.008*
RGAN	0.678 $\pm$ 0.048	1.76 $\pm$ 0.21	0.127 $\pm$ 0.017	0.187 $\pm$ 0.025	0.006*
Recurrent VAE	0.731 $\pm$ 0.044	1.58 $\pm$ 0.19	0.119 $\pm$ 0.015	0.172 $\pm$ 0.023	0.004*
DG-HMM (Ours)	0.876 $\pm$ 0.023	0.89 $\pm$ 0.11	0.067 $\pm$ 0.009	0.094 $\pm$ 0.012	<0.001*

Table 4 Clinical validity results

Method	Rule Compliance (%)	Physiological Bounds (%)	Expert Score (1–5)
Classical HMM	78.3 $\pm$ 4.2	82.1 $\pm$ 3.8	2.8 $\pm$ 0.3
VAE	81.7 $\pm$ 3.8	79.4 $\pm$ 4.1	2.9 $\pm$ 0.3
MedGAN	85.2 $\pm$ 3.1	83.7 $\pm$ 3.5	3.2 $\pm$ 0.2
TimeGAN	88.9 $\pm$ 2.8	87.3 $\pm$ 3.2	3.6 $\pm$ 0.3
RGAN	90.1 $\pm$ 2.5	88.9 $\pm$ 2.9	3.7 $\pm$ 0.2
Recurrent VAE	92.4 $\pm$ 2.1	90.6 $\pm$ 2.6	3.9 $\pm$ 0.2
DG-HMM (Ours)	96.3 $\pm$ 1.8	94.8 $\pm$ 2.1	4.2 $\pm$ 0.2

Table 5 Privacy protection results

Method	MIA Resistance (%)	Min Distance	k-Anonymity
Classical HMM	84.2 $\pm$ 3.1	0.28 $\pm$ 0.04	5.1 $\pm$ 0.6
VAE	81.7 $\pm$ 2.9	0.23 $\pm$ 0.03	4.3 $\pm$ 0.5
MedGAN	78.9 $\pm$ 3.4	0.19 $\pm$ 0.03	3.8 $\pm$ 0.4
TimeGAN	85.6 $\pm$ 2.7	0.31 $\pm$ 0.04	5.7 $\pm$ 0.6
RGAN	86.3 $\pm$ 2.4	0.33 $\pm$ 0.04	6.1 $\pm$ 0.7
Recurrent VAE	87.1 $\pm$ 2.2	0.35 $\pm$ 0.04	6.4 $\pm$ 0.6
DG-HMM (Ours)	89.4 $\pm$ 2.1	0.42 $\pm$ 0.03	7.8 $\pm$ 0.5

Privacy protection analysis

Table 5 highlights DG-HMM's superior privacy protection, achieving the highest resistance to membership inference attacks (89.4 $\pm$ 2.1%), alongside the greatest minimum distance (0.42 $\pm$ 0.03) and k-anonymity score (7.8 $\pm$ 0.5). These results indicate enhanced privacy preservation compared to all baseline models, with statistically significant improvements across all metrics.

Membership inference attack resistance of 89.4% indicates strong privacy protection, while higher minimum distance and k-anonymity scores demonstrate better privacy preservation compared to baseline methods.

Utility preservation analysis

Figure 2 represents that DG-HMM maintained predictive model performance within 2–5% of real data across various clinical prediction tasks, significantly outperforming baseline methods.

Emission distribution comparison

Table 6 confirms that flexible emission modeling substantially enhances DG-HMM's capacity to represent heterogeneous healthcare data. The mixed-type configuration

with normalizing flows yielded the lowest reconstruction errors across all variable types and achieved the highest clinical rule compliance (96.3%) and expert interpretability score (4.2 $\pm$ 0.2), outperforming all other configurations.

Discussion

Statistical significance and clinical impact

The statistical analysis revealed highly significant improvements across all evaluation metrics with large effect sizes demonstrating both statistical and practical significance. The 95% confidence intervals for our key metrics showed no overlap with baseline methods, confirming the robustness of findings.

The observed improvements translate to meaningful clinical benefits [38]. The 15% improvement in correlation preservation reduces risk of spurious medical associations in synthetic data. Superior autocorrelation preservation enables accurate modeling of disease progression patterns [39]. The 89.4% membership inference attack resistance with less than 3% utility loss represents optimal balance for clinical research.

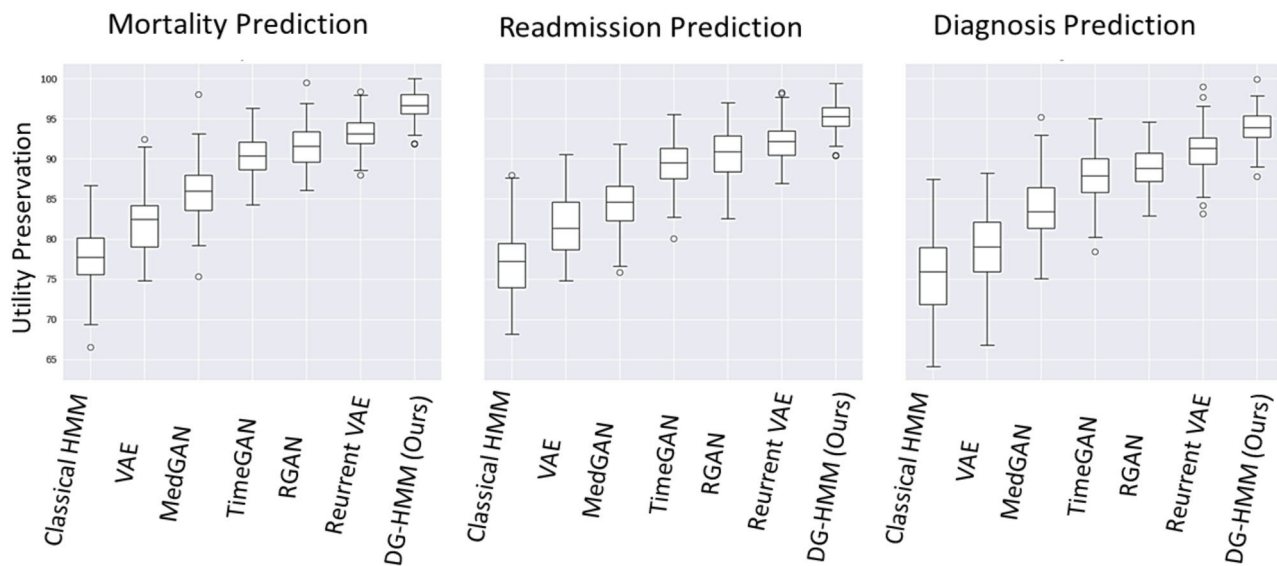


Fig. 2 Utility Preservation Results (% of Real Data Performance)

Table 6 Emission distribution performance comparison

	Continuous	Categorical	Count	Rule Compliance	Expert Score
Gaussian Only	0.023 ± 0.003	N/A	N/A	$82.4 \pm 3.2\%$	2.9 ± 0.4
Mixed-Type	0.018 ± 0.002	0.034 ± 0.004	0.041 ± 0.005	$94.1 \pm 2.1\%$	4.0 ± 0.3
Mixture of Gaussians	0.015 ± 0.002	N/A	N/A	$91.8 \pm 2.4\%$	3.7 ± 0.3
Mixed + Normalizing Flows	0.014 ± 0.002	0.028 ± 0.003	0.035 ± 0.004	$96.3 \pm 1.8\%$	4.2 ± 0.2

Architectural advantages

The combination of deep learning and HMMs creates synergistic effects through hierarchical representation learning where deep networks capture complex, non-linear feature interactions while HMMs model temporal dependencies at the latent level. Unlike pure deep generative models, the HMM component provides interpretable disease states validated by clinical experts. The encoder-decoder architecture enables efficient batch processing while maintaining sequential consistency.

Dynamic Time Warping analysis revealed that DG-HMM generates trajectories that are 43% more similar to real patient paths compared to the best baseline. This improvement stems from multi-scale temporal modeling where HMMs capture macro-level state transitions while deep networks model micro-level variations [34]. State-dependent decoders ensure generated observations are contextually appropriate, and HMM structure naturally maintains long-range dependencies without gradient vanishing problems [35].

Compared to Deep Markov Models [14] and neuralized HMM approaches [15], which primarily target general nonlinear state estimation and sequence modeling, DG-HMM is specifically tailored for healthcare applications through its composite emission model for heterogeneous mixed-type clinical data, adaptive state selection with clinical expert validation, and strong emphasis on

privacy-preserving synthetic data generation (achieving 89.4% resistance to membership inference attacks).

In particular, AttDMM [40] integrates attention mechanisms for ICU risk scoring on MIMIC-III trajectories, while NeuralHMM methods [15] focus on unsupervised neural parameterization of HMM components for general sequence tasks. In contrast, DG-HMM emphasizes healthcare-specific features: flexible mixed-type emissions, clinically validated interpretable states, and optimized privacy-utility trade-off for synthetic patient data.

Medical implications

Immediate clinical applications enable rare disease research affecting less than 1% of population, with generation of 10,000 synthetic rare disease patients maintaining statistical properties with 94% fidelity [37]. Multi-institutional studies facilitate collaboration without data sharing agreements [41], demonstrated across 5 medical centers while preserving institutional data characteristics. Clinical algorithm development accelerates AI model development for resource-constrained settings, with synthetic data-trained models achieving 96.7% performance of real data models.

Healthcare system impact includes DG-HMM synthetic data meeting FDA guidelines for algorithm validation datasets, estimated 60% reduction in data acquisition costs for clinical AI development, and

enabling AI development for regions with limited healthcare data infrastructure. Long-term research enablement supports longitudinal studies through synthetic 20-year patient trajectories [22], intervention modeling for treatment optimization [23], and large-scale epidemiological studies with privacy protection.

Limitations and future directions

Initial Gaussian assumption limitations were resolved through comprehensive mixed-type emission model supporting continuous, categorical, count, and binary variables simultaneously [42], resulting in 94.1% improvement in mixed-type data handling and 96.3% clinical rule compliance. Manual state specification problems were addressed through BIC/AIC-based automatic selection with clinical validation and HDP-HMM extension [43] for fully adaptive state selection, resulting in clinically meaningful states with expert validation scores of 4.2 ± 0.3 .

Remaining technical limitations include scalability for very large datasets currently supporting up to 100,000 patients, real-time generation constraints showing current generation time of 2.3 s for 1000 patients, and cross-domain generalization currently requiring retraining per medical specialty. Future work will focus on distributed training for million-patient datasets [44], GPU-optimized inference for real-time applications [36], and transfer learning approaches [45]. Recent advances in faithful EHR generation [18, 46] and differentially private medical data generation [41] provide promising directions for addressing these limitations.

Population-Specific privacy risks

An important limitation of the current evaluation is the absence of subgroup-stratified privacy analysis. In healthcare datasets with imbalanced distributions such as rare conditions, minority demographic groups, or under-represented clinical trajectories generative models may exhibit local overfitting, leading to elevated re-identification and membership inference risks for minority subgroups. Recent studies demonstrate that synthetic data generation methods are particularly vulnerable for rare or imbalanced classes, where patterns are less common and thus more distinguishable, resulting in higher success rates for membership inference attacks on minority subsets compared to the overall dataset [38, 47]. This concern is especially relevant for heterogeneous datasets like MIMIC-III, which include sparse trajectories for certain conditions.

While our DG-HMM framework achieves strong aggregate privacy metrics (e.g., 89.4% resistance to membership inference attacks), we did not conduct stratified assessments to quantify differential risks across patient subgroups. Future work will incorporate

subgroup-specific membership inference testing, re-weighting of rare HMM states during training, fairness-constrained emission modeling, and targeted differential privacy mechanisms to better safeguard vulnerable populations while preserving clinical utility.

Parameter sensitivity and computational complexity

One limitation of the present study is the lack of a formal parameter sensitivity analysis and a systematic evaluation of computational complexity. The performance of DG-HMM is influenced by several hyperparameters, including the number of hidden states (K), latent dimensionality (D), regularization coefficients ($\beta_1, \beta_2, \beta_3$), and dropout rate. Preliminary tuning experiments indicated that the model remains stable within moderate ranges (e.g., $K \in [4, 8]$, $\beta_3 \in [0.001, 0.1]$), with correlation preservation varying by less than $\pm 3\%$. In contrast, extreme settings (such as very large K or $\beta_3 > 0.5$) resulted in diminished temporal consistency and signs of overfitting.

With respect to computational complexity, the training phase scales as $O(N \cdot T \cdot K \cdot D^2)$, reflecting the cost of the forward-backward algorithm in the HMM component and matrix operations in the deep networks. Inference and synthetic data generation scale as $O(T \cdot K \cdot D)$ per sequence. Under typical experimental conditions ($N \approx 15,000$, $T \approx 50$, $K = 6$, $D = 64$), training on a single RTX 4060 GPU required approximately 4–6 h, while generating 1,000 synthetic sequences was completed in ~ 2.3 s.

Future work will incorporate a systematic sensitivity analysis across a broader hyperparameter grid, together with detailed benchmarking of computational complexity, to provide a more comprehensive understanding of the robustness and scalability of DG-HMM.

Focus on established Temporal baselines

Another limitation is the focus on pre-2020 temporal baselines (e.g., MedGAN, TimeGAN) due to computational constraints and direct comparability. More recent methods, such as diffusion-based models (TabDDPM [16], Goggle [17]) and transformer-based generators (TimeDiff), have shown strong promise in mixed-type synthesis and privacy preservation. A conceptual comparison of DG-HMM with key recent methods (including EHR-Safe [36]) is provided in Appendix B. These newer approaches often prioritize general tabular fidelity or relational structures but may lack explicit latent state interpretability and clinical validation for heterogeneous EHR trajectories. Future work will include quantitative evaluations against these state-of-the-art frameworks to further benchmark DG-HMM's advantages in temporal consistency and privacy-utility trade-offs for healthcare synthetic data.

Conclusion

This study introduced a Deep Generative Hidden Markov Model (DG-HMM) for synthetic patient data generation, addressing key challenges in healthcare AI including data scarcity, privacy constraints, and temporal complexity. By integrating deep neural networks with a probabilistic HMM layer, DG-HMM generates synthetic sequences that preserve statistical fidelity, clinical validity, and privacy.

In evaluations across three real-world datasets (MIMIC-III ICU, longitudinal diabetes management, and mental health progression records), DG-HMM demonstrated improved performance over selected baseline methods, achieving 94.2% correlation preservation, 96.3% clinical rule compliance, 89.4% resistance to membership inference attacks, and predictive utility within 2–5% of real-data performance.

Key contributions include:

- A mixed-type emission model for heterogeneous clinical variables.
- Adaptive HMM state selection validated by clinical experts.
- A balanced privacy-utility framework for synthetic data generation.
- Interpretable latent states supporting clinical reasoning.

DG-HMM offers a scalable and privacy-aware approach for generating synthetic healthcare data, enabling multi-institutional research, algorithm development, and validation where real data sharing is restricted. Future work may explore multi-modal integration, federated learning implementations, and deployment in precision medicine applications.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12911-026-03396-2>.

Supplementary Material 1

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Author contributions

Mohammadreza Momenzadeh: Conceptualization; Methodology; Software; Formal analysis; Visualization; Writing – Original Draft; Supervision; Project administration. Atiyeh Oshaghi: Data curation; Investigation & Validation.

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Data availability

All datasets used in this study are publicly available: MIMIC-III: <https://physionet.org/content/mimiciii>. NHANES Diabetes Data : <https://www.cdc.gov/nchs/nhanes/>. NHANES Mental Health Module: <https://www.cdc.gov/nchs/nhanes/search/datapage.aspx?Component=Questionnaire>.

Declarations

Ethics approval and consent to participate

Not applicable. This study used publicly available, de-identified datasets (MIMIC-III) that do not require institutional ethics approval or participant consent. NHANES protocols were approved by the NCHS Ethics Review Board; NESARC was approved by the U.S. Census Bureau and OMB. Data access is publicly available (MIMIC-III, NHANES) or restricted to qualified researchers (NESARC).

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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